

Signed Forcing and Arithmetic Compensation in the SAM Kernel

Exact transport, source-conditioned bounds, and a conditional route to the Riemann hypothesis

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Abstract

We develop an exact signed-energy framework for stopped Möbius sources in the original SAM kernel. An intrinsic resistance coordinate preserves the full derivative norm and exposes a forced dyadic transport law. Its energy identity separates inherited source energy, complete signed forcing, and a geometric remainder bounded uniformly from above. We derive exact admission identities, an arbitrary-length conditional bound for equal-source admission runs, and a conditional route from subpower forcing excess to the Riemann hypothesis. A further reduction shows that the required smaller-source transfer may have an energy exponent approaching two: a saving of order $1/\log \log s$ is already sufficient. Exact native computations cover 12,836 stopped prefixes at four dyadic scales. The largest excess is 0.520439137454; the candidate ceiling one survives this scope, while the half-ceiling has counterexamples. Source histories resolve both forcing-led buildup and a peak admission caused entirely by decreasing inherited energy. The uniform arithmetic estimate remains open. The contribution is a collection of exact identities, conditional bounds, computational evidence, and an explicit sufficient threshold for that estimate.

Keywords: Möbius function; Riemann hypothesis; signed energy; dyadic transport; arithmetic compensation; source-conditioned estimates.

1. The research question

The central question is whether the complete signed interaction of the actual Möbius source admits a uniform subpower bound. Our investigations retain the original kernel, its mean term, its weighted cuts, and the placement of each positive part in the original SAM formulation. This paper studies the full energy that bounds the retained interaction and the signed excess governing its transport between dyadic scales.

The computational witnesses make cancellation visible at several levels. Large source contributions can

compensate across divisor bands, an excursion can be repaid several admissions later, and a local excess can rise even when the transported source receives no net admission. These distinctions motivate an exact accounting of the signed history.

Our principal results are the transport identity and its uniform geometric remainder, the equal-admission run bound, and two conditional implications: subpower excess implies RH, and a sufficiently small saving below quadratic smaller-source transfer implies subpower excess. We distinguish those derived implications from their remaining arithmetic hypotheses throughout.

2. The original energy and its intrinsic coordinate

Let s be a positive dyadic integer and $0 \leq t \leq s$. The stopped source is

$$\begin{aligned} a_i &= \mu(s+i) \mathbf{1}_{\{i < t\}}, & 0 \leq i < s. \\ M_a &= \sum_i a_i, \\ q_a(r) &= \sum_{\{i < r\}} a_i - r M_a/s, & 0 \leq r \leq s. \end{aligned}$$

Thus $q_a(0)=q_a(s)=0$. The full energy and associated inner product are

$$\begin{aligned} Q_s(a) &= M_a^2/s + \sum_{r=1}^{s-1} q_a(r)^2/[r(s-r)], \\ \langle a, b \rangle_s &= M_a M_b/s + \sum_{r=1}^{s-1} q_a(r)q_b(r)/[r(s-r)]. \end{aligned} \quad (1)$$

Write K_s for this positive definite kernel. Its inverse quadratic form is

$$\begin{aligned} R_s(f) &= (\sum_i f_i)^2/s \\ &\quad + \sum_{r=1}^{s-1} r(s-r) (f_{r-1} - f_r)^2. \end{aligned} \quad (2)$$

The constant mode has kernel eigenvalue one; the remaining eigenvalues are $1/[k(k+1)]$, for $1 \leq k < s$. The companion kernel derivation and its exact native qualification are recorded in [1].

For $s \geq 2$ introduce the resistance coordinate

$$y_0 = 0, \quad y_i = \sum_{r=1}^{i-1} 1/[r(s-r)].$$

If F is linear between the nodes (y_i, f_i) , and $\bar{f}$ is the arithmetic mean of the node values, then

$$\begin{aligned} R_s(f) &= s \bar{f}^2 + \int |F'(y)|^2 dy, \\ \sum_i a_i f_i &= M_a \bar{f} - \int q_a(y) F'(y) dy, \end{aligned} \quad (3)$$

where $q_a(y)=q_a(r)$ on the edge from y_{r-1} to y_r . Both integrals run from y_0 to y_{s-1} . The mean is the node mean. For $t \geq 1$, the equivalent stopped-endpoint identity is

$$\begin{aligned} \sum_{i < t} a_i f_i &= M_a f_{t-1} \\ &- \sum_{r=1}^{t-1} (\sum_{i < r} a_i) (f_r - f_{r-1}). \end{aligned} \quad (4)$$

Equations (3) and (4) follow by finite summation by parts. They preserve the endpoint and derivative terms needed by the complete source functional.

3. Exact dyadic forcing and the geometric remainder

Define the actual stopped fine source and the parity forcing by

$$\begin{aligned} z_j &= \mu(2s+j) \, 1_{\{j < 2t\}}, \\ b_i &= [1_{\{s+i \text{ even}\}} \mu(s+i) + \mu(2(s+i)+1)] \, 1_{\{i < t\}}. \end{aligned}$$

The identity $\mu(2n) = -\mu(n)$ for odd n , and $\mu(2n) = 0$ for even n , gives

$$\begin{aligned} v_i &:= z_{2i} + z_{2i+1} = -a_i + b_i, \\ d_i &:= z_{2i} - z_{2i+1}. \end{aligned}$$

If $c_r = q_v(r)$, the fine cuts satisfy

$$\begin{aligned} q_z(2r) &= c_r, \\ q_z(2i+1) &= (c_i + c_{i+1} + d_i)/2. \end{aligned} \quad (5)$$

Proposition 1 — Full signed transport. With

$$\begin{aligned} A(s, t) &= Q_s(b)/2 - \langle a, b \rangle_s, \\ E(s, t) &= A(s, t) - Q_s(a)/2, \end{aligned}$$

the original energy satisfies

$$Q(2s, 2t) = Q(s, t) + E(s, t) + R(s, t), \quad R(s, t) \leq 3. \quad (6)$$

Here $Q(s, t)$ denotes the energy of the stopped Möbius source at (s, t) . The remainder is one-sided; it is not assigned an absolute bound of three.

Derivation. Put $h_r = r(s-r)$ and $\omega_i = 1/[(2i+1)(2s-2i-1)]$. Substitution of (5) in (1) gives

$$\begin{aligned} S &= \tfrac{1}{4} [\sum_{r=1}^{s-1} c_r^2 / h_r - \sum_{i=0}^{s-1} \omega_i (c_i + c_{i+1})^2], \\ I &= \tfrac{1}{2} \sum_i \omega_i (c_i + c_{i+1}) d_i, \end{aligned}$$

$$\varepsilon = \frac{1}{4} \sum_i \omega_i d_i^2,$$

$$R = -S + I + \varepsilon. \quad (7)$$

The interpolation loss has an explicit nonnegative decomposition. With $g_r = c_r/h_r$ for interior r and $g_0 = g_s = 0$,

$$S = \frac{1}{8} \sum_{r=1}^{s-1} (\omega_{r-1} + \omega_r) c_r^2 / h_r + \frac{1}{4} \sum_{i=0}^{s-1} \omega_i h_i h_{i+1} (g_i - g_{i+1})^2 \geq 0. \quad (8)$$

Let $\Omega = \sum_i \omega_i = (H_{2s} - H_s/2)/s$, where H_n is the harmonic number. Then $\Omega \leq 1$. The coefficient bounds imply $|d_i| \leq 2$ and $|c_r| \leq 4h_r/s$. Since $2(h_i + h_{i+1}) + 1 = 1/\omega_i$, they give $\varepsilon \leq \Omega$ and $|I| \leq 2(1 - \Omega/s)$. Consequently $R \leq 3$. Polarization gives $Q_s(v)/2 = Q_s(a)/2 + A$, completing (6).

An odd fine stop adds the exact signed increment $J(s, t) = Q(2s, 2t+1) - Q(2s, 2t)$. Positivity and unit row sums of K imply $J \leq 2$. For completeness, its entries for $i \leq j$ are

$$(K_s)_{ij} = 1/s^2 + [2H_{s-1} - H_j - H_{s-1-i}]/s.$$

They are nonnegative; the row-sum statement also follows directly from (1). At a fresh admission, the previously zero coordinate gives an energy increment at most $2\sum_{i < j} K_{ji} + K_{jj} \leq 2$. Thus, whenever the stop is admissible,

$$Q(2s, 2t + \varepsilon_0) = Q(s, t) + E(s, t) + R(s, t) + \varepsilon_0 J(s, t), \quad \varepsilon_0 \in \{0, 1\}. \quad (9)$$

4. Source-conditioned excess and arithmetic compensation

At the next admission $j = t$, retain the current vectors and write $\alpha = \mu(s+j)$, $\beta = 1_{\{s+j \text{ even}\}} \mu(s+j) + \mu(2(s+j)+1)$, $p = (K_s a)_j$, $q = (K_s b)_j$, and $k = (K_s)_{jj}$. Expanding the quadratic forms gives the exact signed admission identity

$$\Delta E = (\beta - \alpha)q - (\beta + \alpha)p + (\beta^2/2 - \alpha\beta - \alpha^2/2)k. \quad (10)$$

Its three separately retained contributions are

$$\begin{aligned} \Delta(Q_b/2) &= \beta q + \beta^2 k/2, \\ \Delta(-(a, b)) &= -\alpha q - \beta p - \alpha \beta k, \\ \Delta(-Q_a/2) &= -\alpha p - \alpha^2 k/2. \end{aligned} \quad (11)$$

Proposition 2 — Equal-admission run bound. Suppose every admission in a consecutive source interval satisfies $\beta = \alpha$. At every stop t in that interval,

$$E(t) - E(\text{start}) = Q_s(a_{\text{start}}) - Q_s(a_t) \leq Q_s(a_{\text{start}}). \quad (12)$$

Derivation. Each admission leaves $v=b-a$ unchanged. The exact identity $E=Q_s(v)/2-Q_s(a)$ therefore telescopes through the entire interval. Nonnegativity of the terminal parent energy gives the upper bound.

This bound is independent of the run length and permits nonmonotone parent energy. Its hypothesis specifies the arithmetic family to which it applies. It does not assert that later admissions will continue to satisfy $\beta=\alpha$.

The new scale-8192 peak illustrates the mechanism. At stop 7798, the admitted parent 15989 and odd child 31979 both have Möbius value +1. Hence $\alpha=\beta=1$: the pair source v receives zero net admission, and the excess rises entirely through decreasing parent energy. The next admission has $\alpha=-1$ and $\beta=-2$, and all three contributions in (11) are negative.

Admission	Forcing self	Forcing cross	Inherited	Total ΔE
8192,7798	+0.013340219	-0.005610452	+0.007729767	+0.015459533
8192,7799	-0.026301827	-0.002307442	-0.007729178	-0.036338447

Across the larger interval 7734–7798, forcing contributes +0.153780483399 and the inherited contribution is -0.020993960975. Thus the full buildup is forcing-led, while its final peak admission has the equal-admission mechanism. The first later return to the 7734 excess baseline occurs at 7815. Signed drifts at 4, 8, 32, and 64 admissions after the peak are all negative. These observed outcomes and the conditional bound (12) have distinct scopes.

5. Subpower excess and the conditional implication to RH

Theorem 3 — Conditional SAM-to-RH implication. Suppose that for every $\eta>0$ there is a finite C_η , independent of the dyadic scale and stop, such that

$$E(s, t) \leq C_\eta s^\eta, \quad s \geq 2 \text{ dyadic}, \quad 0 \leq t \leq s. \quad (13)$$

Then the full SAM energy is uniformly subpower, and RH follows.

Derivation. Let $H_j = \max_t Q(2^j, t)$. Equation (9) gives $H_{j+1} \leq H_j + C_\eta 2^{j\eta} + 5$. Therefore

$$H_K \leq H_1 + C_\eta (2^{K\eta} - 2^\eta) / (2^\eta - 1) + 5(K-1). \quad (14)$$

Using a smaller positive exponent to absorb the logarithmic term shows $H_K = O_\delta(2^{K\delta})$ for every $\delta>0$. Since the cut energy is nonnegative,

$$|\sum_{i<t} \mu(s+i)| \leq \sqrt{s Q(s, t)}. \quad (15)$$

Partition an initial integer interval into complete dyadic blocks and one stopped block. The geometric sum of the bounds (15) yields $M(x) = \sum_{n \leq x} \mu(n) = O_\epsilon(x^{1/2+\epsilon})$ for every $\epsilon>0$.

For $\text{Re}(w)>1/2$, define $F(w) = w \int_1^\infty M(x) x^{-w-1} dx$. The preceding estimate gives locally uniform convergence, including differentiated integrals, so F is holomorphic in that half-plane. Partial summation and the classical Möbius Dirichlet series [7] give $F(w) = 1/\zeta(w)$ for $\text{Re}(w)>1$. The identity theorem implies $\zeta(w)F(w) = 1$ throughout the connected half-plane with its pole at 1 removed. There are consequently no zeros to

the right of the critical line. The zeta reflection formula [8] excludes nontrivial zeros to its left.

The implication uses hypothesis (13) with all its quantifiers. Establishing that uniform actual-source estimate remains the central arithmetic task.

Signed compensation is retained even before taking the bound (14). For a terminal source $(2^K, T)$, set $t_j = \lfloor T/2^{K-j} \rfloor$ and $\varepsilon_j = t_{j+1} - 2t_j$. Then

$$Q(2^K, T) = Q(2, t_1) + \sum_{j=1}^{K-1} [E(2^j, t_j) + R(2^j, t_j) + \varepsilon_j J(2^j, t_j)]. \quad (16)$$

Every negative remainder and endpoint increment remains in this exact sum.

There is also a useful reverse energy estimate. The pair source has the same mean as z and $q_v(r) = q_z(2r)$; direct comparison of the nonnegative terms in (1) gives $Q_s(v) \leq 4Q_{2s}(z)$. Hence

$$E(s, t) \leq 2Q(2s, 2t) - Q(s, t). \quad (17)$$

Together with (14), this establishes equivalence between one-sided subpower excess and uniform subpower full energy. Equation (17) itself is not an upper induction for the fine energy.

6. A smaller-source transfer threshold approaching the critical power

Let $u = \lfloor \sqrt{2s-1} \rfloor$. The complete source functional has the exact divisor form

$$L_{\{s, t\}}(f) = -\sum_{\{a, b \leq u\}} \mu(a)\mu(b) \sum_{\{v = \lfloor s/(ab) \rfloor\}^{\lfloor (s+t-1)/(ab) \rfloor}} f(abv-s). \quad (18)$$

Let U_s be the sum of original full dyadic energies of $\mu(1), \dots, \mu(u)$, with the last block zero-padded. It includes the $\mu(1)$ block, so $U_s \geq 1$. Let H_m be the first $m+1$ kernel modes, $m = \min(s-1, \lfloor \sqrt{s} \rfloor)$, and F_m the energy in those modes. The original spectral calculation gives

$$F_m = \sup_{\{f \in H_m, f \neq 0\}} |L_{\{s, t\}}(f)|^2 / R_s(f), \\ F_m \leq Q(s, t) \leq F_{m+1}. \quad (19)$$

Theorem 4 — Vanishing saving below quadratic transfer. Suppose, for $s = 2^j$, $j \geq 1$, all actual stops and all $f \in H_m$, that

$$|L_{\{s, t\}}(f)|^2 \leq C(1+j)^A U_s^{p_j} R_s(f), \\ p_j = 2 - \kappa / \log(e+j), \quad 0 < \kappa \leq 1, \quad C, A \geq 0 \text{ fixed.} \quad (20)$$

Then (13) holds. More explicitly, for each $0 < c < \kappa / (2 \log 2)$, suitable constants C_0, C_1 give

$$E(s,t) \leq C_0 \exp\{C_1 \log(2s)/[\log \log(e^e+2s)]^c\}. \tag{21}$$

Derivation. Put $J_n=1+\sum_{j=0}^n H_j$. The largest dyadic input block in $U_{\{2^j\}}$ has index $\lfloor j/2 \rfloor$, including the exact endpoint subtraction in u . Thus $U_{\{2^j\}} \leq J_{\lfloor j/2 \rfloor}$. Summing (19) and (20), and using the monotonicity of p_j , gives a fixed $B \geq 1$ such that

$$J_n \leq B(n+1)^{A+1} J_{\lfloor n/2 \rfloor}^{p_n}. \tag{22}$$

Set $g_m=2^{\{-m\}\log J_{\{2^m\}}}$. Taking logarithms yields

$$g_m \leq [1-\kappa/(2\log(e+2^m))]g_{m-1}+O((m+1)/2^m).$$

For any c in the stated range, the bracket is eventually at most $1-c/m$. Products of these coefficients from $r+1$ through m are $O((r/m)^c)$. The convergent sum $\sum_r r^c(r+1)/2^r$ then gives $g_m=O(m^{-c})$. Monotonicity extends this to $\log J_n=O(n/(\log n)^c)$. Equation (17) supplies (21).

For a fixed exponent $1 < p < 2$, the same argument gives the simpler envelope $E(s,t) \leq \exp\{O((\log s)^{\log_2 p})\}$. At $p=1$, it recovers $\log J_n=O((\log(n+1))^2)$. At exactly $p=2$, recurrence (22) permits $\log J_n$ of order n ; a strict saving has mathematical content.

Theorem 4 identifies a sufficient threshold within the existing source functional. Hypothesis (20) is open. The actual Möbius factors and their signed coupling are essential to the source question; no unrestricted bilinear operator estimate is substituted for it.

7. Computational findings and verification scope

The complete forcing-excess scans give the following results. Counts include both endpoints of each stopped-prefix range.

Scale s	Prefixes	Maximum excess	Maximizing stop	Failures of $E \leq 1/2$
32	33	0.063413987429	1	0
512	513	0.210217371429	481	0
4096	4097	0.514400059116	1071	5
8192	8193	0.520439137454	7798	8
Total	12,836			13

The candidate $E \leq 1$ survives every prefix in this scope. **The test result suggests the concept is possible.** For the uniform half-ceiling candidate, the exact counterexamples give: **The test falsifies the concept.** These classifications concern the declared constant candidates and their scope.

The complete signed divisor campaign separately covers 15,364 prefixes at 1024–8192 for its retained-energy-to-input-energy gain. These counts refer to a different quantity and are not added to the excess coverage. Its signed band data illustrate the scale of compensation: at (8192,6994), band self energies sum to approximately

28,252.5604, their signed interband cross terms to $-28,252.1849$, and the complete retained energy is about 0.37548 [2].

The scale-8192 forcing scan takes 186.53 native CPU seconds. Its selected witnesses and two subsequent source refinements supply 122,915 native internal checks, with 169 independent exact checks of energy, source transitions, and the signed interval. A separate signed-ancestry study supplies 48,340 native and 356 independent checks. These are scoped identity checks, not independent random samples and not a verification of an unbounded arithmetic hypothesis.

Arithmetic runs through the source-bound GEN3 native C++/GMP adapter under the current SAM domain session. Exact rational outputs, code bindings, source hashes, and execution receipts are preserved. Python orchestration and independent rational recombinations are identified separately.

Selection provenance is retained. The 8192 scan is fixed complete-scale exploration; its two later source calls refine the newly found maximum. Neither is presented as a learned selection. Whole-scale 4096 is excluded from fitting, and the new 8192 results in this study also remain outside fitting. Frozen sign-class forecasts from the earlier source study failed their declared future bounds; conditional full-state bounds have a distinct mathematical basis [4].

Current-state certificates and the selection audit

The earlier scale-4096 study also establishes conditional four-admission bounds from the current full weighted source state. At each next integer index, squareful parent and odd-child coefficients are zero; squarefree coefficients are allowed either sign. GEN3 retains every such branch, updates the exact signed state along each path, and records the maximum excess and signed terminal-drift interval. Future nonzero Möbius signs are not inputs to this construction. Dependencies among their signs are relaxed, so every actual continuation is contained in the enumerated set.

Starting state at scale4096	Stop	Certified upper E, next four admissions
Peak	1071	0.589424758671
Threshold recrossing	1077	0.600499614290
Recovery	1130	0.343086069882

The recovery state therefore certifies E below one-half through the next four admissions. Its terminal-drift interval is approximately $[-0.072035796, +0.081854948]$, explicitly allowing either sign of drift. The three certificates all stay below one over their stated horizon. The recrossing certificate alone does not certify staying below one-half.

Their validity is a finite induction: every admissible actual next pair lies in an enumerated branch, and the exact state update propagates inclusion through the four admissions. **The test result suggests strong contact with the concept.** Scope: these conditional current-state bounds. A state condition that renews such control through continuing admissions remains open.

The frozen sign-pattern predictor is a separate object. Its fitted bounds failed on later scale512 windows,

separate scale2048, and held-out scale4096. For those declared predictive bounds: **The test falsifies the concept.** The current-state certificates use weighted magnitudes and complete signed continuations; they do not use fitted class maxima as upper bounds.

The frozen selection audit records24 learned questions at2.059058 native CPU seconds and two fixed questions at0.134178 seconds. These are combined original derivative-plus-forcing call costs, not isolated marginal costs of forcing. Neither arm improves the known scale512 excess maximum. Their signed transition outcomes and coverage contributions are recorded separately. Different question counts and budgets do not establish equal-budget superiority. These historical audit outcomes do not include the subsequent scale8192 full scan reported above. The target and held-out fitting boundary remain unchanged [4].

8. What the work establishes and what remains

The framework gives an exact connection between arithmetic admissions, inherited energy, and signed forcing at the next scale. Its geometric loss and odd endpoint are controlled, while the source itself remains explicit. The equal-admission identity supplies a conditional overshoot bound through an arbitrary-length arithmetic run. The larger witness shows that this mechanism occurs inside an actual demanding Möbius window.

The conditional closure argument now has a precise sufficient input, and the transfer theorem permits a saving that decreases with scale. The remaining task is to derive that saving, or a sufficient bound on the complete signed excess history, uniformly for the actual source. Neither a constant empirical ceiling nor observed later compensation is used as that missing deduction.

The work provides a concrete research foundation: exact transport, explicit source mechanisms, reproducible demanding cases, and a quantified arithmetic threshold. These results make the continuing uniform-estimate problem more specific while preserving the mathematical structure that produced them.

Contributions and research record

Sean Brady: originator and conceptual director of SAM/SLC; research direction, source questions, campaign priorities, and the decision to pursue the signed excess and its arithmetic causes.

OpenAI ChatGPT: collaborative mathematical development and the supplied original-kernel and dyadic forcing-energy handoffs, including the full-energy transport decomposition and geometric remainder analysis.

Codex: source-bound implementation, exact admission and source-run deductions, native experiment orchestration, independent rational checks, conditional analytic continuation, transfer-threshold derivation, and preparation of this manuscript.

GEN3/SLC: computational execution and recorded learning/selection within the declared source-bound operations. Computational evidence is attributed to the operations and receipts that produced it.

Data and manuscript availability

The verified research archive SAM_RH_UPDATE_2026-09-14_9c42418b3608.zip contains the companion derivations,

native results, signed histories, and implementation sources. Its SHA-256 is

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49ed7c85295046f689900ec9ac3c93d87f4d4b8a2045b2c25c57b719490069d3
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The archive's 290 payload files have individual hashes. This paper is a subsequent synthesis supplied separately. The mathematical history is recorded in entries H001420, H001423–H001426, H001428, H001430, and H001431. The focused archive transfer is recorded in H001432. Full runtime recovery uses the separately retained engine base and targeted delta.

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